* OTHER INFO *

P3 Plug (Single Penetration)....................T3487KX
P3 Plug (2nd Penetration).......................T3487KY
Top Fill Adapter............................46-318197G1
LHe Meter Calibration Sticker...................2100640

*SHIMMING SPECIFICATIONS *

Mechanical Plotting

C6 (40cm x 30cm cylinder)....................< 6.3ppm
40cm DSV (Compact / 0.5T AS).............< 10 ppm
45cm DSV (Sphere)................................< 12 ppm
40cm x 35cm (Elliptical Volume)..........< 45 ppm
45cm DSV (316 point Sphere Cx only)..< 12 ppm

CSI Shimming

20cm DSV (BW 200)................................< 40 Hz
All expansion coefficient..........................≤ ±6 Hz

Quick Shimming

22cm DSV (BW 200)..................................≤ 6 Hz
22cm DSV (GE S-1 / Oxfords)...................≤ 7 Hz
Expansion Coefficients
X3, Y3, Z5, and Z6..................................≤ ±8 Hz
ZXY, Z(X2-Y2), Z3, Z4, Z2X, and Z2Y...≤ ±6 Hz
X, Y, Z1, Z2, ZX, ZY, X2-Y2, and XY.....≤ ±3 Hz

LVshim

45cm DSV (BW 500)................................≤ 45 Hz
22cm DSV (BW 200)................................≤ 6 Hz
Expansion Coefficients Over 45cm DSV
Z1, X, and Y...≤ ±10 Hz
Z2, ZX, ZY, X2-Y2, and XY..................≤ ±25 Hz
X3, Y3, Z3, Z4, Z5, XYZ, Z(X2-Y2), Z2X, and Z2Y................................< 100mA Delta Current /\\
Z6...≤ ± 200 Hz

LVshim Cx & LCC Magnets

45cm DSV (BW 500 1.5T).........................≤ 50 Hz
45cm DSV (BW 500 1.0T).........................≤ 33 Hz
22cm DSV (BW 200)................................≤ 4 Hz
Expansion Coefficients Over 45cm DSV
Z1, X, and Y...≤ ±10 Hz
Z2, Z3, ZX, ZY, X2-Y2, and XY, Z2Y, Z2X, Z(X2-Y2), ZXY, Y3, and X3.................≤ ±25 Hz
Z4, Z5, and Z6..≤ ± 50 Hz

LVshim CRM 1.5T

40cm DSV (BW 500)................................≤ 36 Hz
22cm DSV (BW 200)................................≤ 6 Hz
Expansion Coefficients Over 40cm DSV
Z1, X, and Y...≤ ±10 Hz
Z2, ZX, ZY, X2-Y2, and XY..................≤ ±18 Hz
X3, Y3, Z3, Z4, Z5, XYZ, Z(X2-Y2), Z2X, and Z2Y................................< 100mA Delta Current /\\
Z6...≤ ± 160 Hz

LVshim Cx & LCC Magnets CRM 1.5T

40cm DSV (BW 500)...............................< 40 Hz
22cm DSV (BW 200)...............................≤ 4 Hz
Expansion Coefficients Over 40cm DSV
Z1, X, and Y...≤ ±10 Hz
Z2, Z3, ZX, ZY, X2-Y2, and XY, Z2Y, Z2X, Z(X2-Y2), ZXY, Y3, and X3.................≤ ±25 Hz
Z4, Z5, and Z6..≤ ± 50 Hz

Magnet Frequency Bandwidth

0.2T	2005 gauss	± 9 gauss
	8.540 MHz	± 40 KHz
0.5T	5000 gauss	± 2.5 gauss
	21.288 MHz	± 10.644 KHz
1.0T	10,025 gauss	± 5 gauss
	42.682 MHz	± 21.288 KHz
1.5T	15,000 gauss	± 7.5 gauss
	63.864 MHz	± 31.932 KHz

* TOOLS *

Ramp and Shim Tool Kits

Main Power Supply 750amp........46-260776G3
Main Power Supply 1000amp......46-260776G4
Shim Power Supply.....................46-260777G3
Shim Lead Cable Kit.......................2135558
Magnet Ramp Equip. Kit...........46-260703G3
Ramp Cables and Holder 750amp........2135435
Ramp Cables and Holder 1000amp.....2180589
Magnetometer.....................................46-251865G4
Auto Data Collection Kit.............46-323012G1
Power Supply Checkout Kit.................2101360
Torque Wrench (Magnet Bolting)...46-268521G1

Coldhead Maintenance Kits

Coldhead/Compressor Main. Kit.....46-281088G3
Shield Cooler Vacuum Pump Kit...46-294047G1
Balzer Repair Kit (Heat Gun).....46-306830G13
Low Pres. Regulator and Hose.....46-306734G1
Lakeshore 208 Thermometer Kit...46-301477G1
Balzer Shield Slide Kit......................46-317693G1
UV light Kit...............................46-318784G2
Portable Temperature Monitor RUO..2171219
Coldhead Insertion Tool VMXII.........2182971
Plexiglass Heat Exch Cover Plate VMXII..2132223
Leak Detector (Snoop)................46-252065P71
Adapter Cable for Coldhead Sleeve.......2136506

* MAGNET MONITORING *

Magnet Monitoring Hardware Parts

Magnet Monitoring Interface Box............T3487PX
Water Flow & Temperature Sensor..........T3487PY
Pressure Transducer...........................46-320840P1
Power Supply (Interface Box)...........46-328516P1
Leybold Compressor Adapter Cable..........2116875
Coldhead Cable (Coldhead Sleeve)?46-318042P1

EDM-4

Modem Telebit Trailblazer......................T3901AA
Modem Worldblazer...............................T3901AG
4.x MR / CT EDM Insite Kit............T3901AD 5.x
EDM Kit..T3901MM 4.6
Software Disk...2110500-4
Block style battery...................................2145430
Magnet Monitor (SHeM)..........................T3504DR

Miscellaneous

Coldhead Cable (Coldhead Sleeve)....46-318042P1
Fill Key..2147002
Desiccant Bag (Moisture Adsorbent).....2135979

Magstat Settings

	Low Limit	High Limit
First Stage	25k	65k
Second Stage	2k	20k
Vessel Pressure	0.10psi	2.25psi
Helium Level	*50.0%	110.0%

*Set minimum level appropriately for your style magnet and configuration.

Water Flow	1.75gpm	4.50gpm
Water Temperature	40.0deg	75.0deg
**Shim Lead Flow	0.1SCFH	3.0SCFH
**Pen. Lead Flow	0.1SCFH	4.0SCFH

**Only on preliminary magnet monitoring sites, most sites will not have this.

LHe Level Trend	0.5%	5.0%

	Data Acq. Chan.	Coef. Table
Fisrt Stage	4	5
Second Stage	5	5
Vessel Pressure	2	4
Helium Level	6	3
Water Flow	7	1
Water Temperature	8	2
*Shim Lead Flow	0	1
*Pen. Lead Flow	1	1
LHe Level Trend	4	4

-LHe Trend alarm is Disable at all times.

* MAGNET PERFORMANCE *

Oxford Magnets (41xxx), (42xxx), & (43xxx)

LHe Boiloff Rate...............................< 0.5 liters/hr
LN2 Boiloff Rate.............................< 2.0 Liters/hr
Min. LHe Fill Level............................40% (Fixed)
Min. LHe Fill Level..........................45% (Mobile)
Min. LHe Ramp Level.........................80% (Fixed)
Min. LHe Ramp Level.......................45% (Mobile)

S-I Magnets (67xxx)

LHe Boiloff Rate...............................< 0.5 liters/hr
LN2 Boiloff Rate.............................< 1.0 Liters/hr
Total Helium Flow Rate...........................7.0 SCFH
Stack..........1.7 SCFH, Fill Port..............0.3 SCFH
Flange.........3.1 SCFH, 40K....................1.8 SCFH
Magnet with 2nd Penetration...................0.1 SCFH

S-I Magnets (67xxx) cont

Min. LHe Fill Level...50%
Min. LHe Ramp Level.......................................75%
Min. LHe Shim Level.......................................50%

S-II(Cxxx) / S-III(Dxxx) / Max(Axxx) Magnets

LHe Boiloff Rate..............................< 0.2 liters/hr
He Vessel Pressure..........................0.25 to 0.5 psig
F1 Flow..0.4 to 0.6 SCFH
F2 Flow..1.5 to 2.0 SCFH
1st Stage...40 to 58 Kelvin
2nd Stage...7 to 17 Kelvin
Min. LHe Fill Level.................30% (3 Dewar Fills)
Min. LHe Fill Level.................50% (2 Dewar Fills)
Min. LHe Fill Level.............50% (YMS Max Site)
Min. LHe Ramp Level.......................................75%
Min. LHe Shim Level.............................50%

S-IV(Gxxx) / SX(Xxxx) Magnets

LHe Boiloff Rate..............................< 0.2 liters/hr
He Vessel Pressure..........................0.25 to 0.5 psig
F1 Flow..1.8 to 2.2 SCFH
F2 Flow..1.3 to 1.7 SCFH
1st Stage...40 to 58 Kelvin
2nd Stage...7 to 17 Kelvin
Min. LHe Fill Level.........................65% (S-IV)
Min. LHe Fill Level...........................50% (SX)
Min. LHe Ramp Level.........................90% (S-IV)
Min. LHe Ramp Level...........................80% (S-X)
Min. LHe Shim Level..........................70% (S-IV)
Min. LHe Shim Level...........................50% (SX)

SV(Jxxx) / SX-C1(Kxxx) / SX-C2(Lxxx) Magnet

LHe Boiloff Rate..............................< 0.2 liters/hr
He Vessel Pressure..........................0.25 to 0.5 psig
F1 Flow..1.8 to 2.2 SCFH
F2 Flow..1.2 to 1.6 SCFH
1st Stage...32 to 60 Kelvin
2nd Stage...7 to 17 Kelvin
Min. LHe Fill Level...50%
Min. LHe Ramp Level.......................................85%
Min. LHe Shim Level.......................................65%

Conquest Cx1.0T(Pxxx) / 1.5T(Nxxx) Magnets

LHe Boiloff Rate.............................< 0.18 liters/hr
He Vessel Pressure..........................0.25 to 0.5 psig
F1 Flow..1.8 to 2.2 SCFH
F2 Flow..0.8 to 1.2 SCFH
1st Stage...32 to 60 Kelvin
2nd Stage...7 to 17 Kelvin
Min. LHe Fill Level...42%
Min. LHe Ramp Level.......................................90%
Min. LHe Shim Level.......................................75%

Compact(Exxx) / 0.5T VMX(Vxxx) Magnets

LHe Boiloff Rate...............< 0.1 liters/hr (compact)
LHe Boiloff Rate...........< 0.075 liters/hr (0.5T AS)
He Vessel Pressure..........................0.25 to 0.5 psig
F1 Flow..0.9 to 1.1 SCFH

Compact(Exxx) / 0.5T VMX(Vxxx) Magnet cont

1st Stage...32 to 60 Kelvin

2nd Stage..7 to 17 Kelvin
Min. LHe Fill Level........................40% (Compact)
Min. LHe Fill Level.........................50% (0.5T AS)
Min. LHe Ramp Level....................75% (Compact)
Min. LHe Ramp Level.....................85% (0.5T AS)

0.5T VMX2(Zxxx) Magnets

He Vessel Pressure.........>0.3 psig but <5 psig
Shield Temp.......................................< 55 Kelvin
Cryocooler 2nd stage Temp........> 3.8K but < 4.6K
Recondenser Temp.....................> 3.8K but < 5K /\\
Temp (Coldhead - Recondensor)..< 0 but < 0.3K
Min. LHe Fill Level..55%
Min. LHe Ramp Level......................................85%

Profile Magnets

Magnet Temperature........................31 to 34 deg. C
Temperature Restrictions.................< 45 deg. C

1.5T LCC (Rxxx) / 1.0T LCC (Qxxx) Magnets

He Vessel Pressure.............>3 psig but <5 psig
Shield Temp.......................................< 55 Kelvin
Cryocooler 2nd stage Temp..........> 3.8K but < 4.6K
Recondenser Temp..............> 3.8K but < 5K /\\
Temp (Coldhead - Recondensor)..< 0 but < 0.3K
F1 Flow..1.8 to 2.2 SCFH
Min. LHe Fill Level..50%
Min. LHe Ramp Level...............................90%
Min. LHe Shim Level.....................................75%

* MAGNET DIODE CURVES *

Magnets Using SI-410 Diodes:
SV, SXC-1, SXC-2, VMX, Cx1.0, Cx1.5, VMX2
***Lakeshore 208 use Curve 6**

SI-410 Diodes Curve

Voltage	Temp. K	Voltage	Temp. K
0.618	273	1.053	60
0.714	230	1.059	56
0.791	195	1.064	53
0.855	165	1.069	50
0.906	140	1.073	48
0.955	115	1.077	45
0.992	95	1.081	43
0.996	94	1.085	40
0.998	93	1.09	37
1	92	1.094	34
1.002	91	1.101	30
1.004	90	1.104	28
1.006	89	1.109	26
1.008	88	1.114	24
1.01	87	1.119	23
1.012	86	1.129	22
1.014	84	1.15	21
1.016	83	1.17	20
1.018	81	1.21	18
1.02	80	1.25	16
1.022	79	1.27	15
1.024	77	1.29	14
1.028	76	1.314	13
1.03	75	1.335	12
1.033	74	1.365	11
1.035	73	1.395	10
1.037	72	1.43	9
1.039	70	1.47	7
1.041	69	1.612	5
1.043	68	1.653	4
1.045	66	1.685	3
1.047	65	1.708	2
1.049	63	1.714	1
1.051	61		

Magnets Using DT-500 Diodes:
SII, SIII, MAX, SIV, Compact
*Lakeshore 208 use Curve 2

DT 500 Diode Curve

Voltage	Temp. K	Voltage	Temp. K
0.247	345	1.093	33
0.364	305	1.097	32
0.42	285	1.1	31
0.474	265	1.105	30
0.539	240	1.111	29
0.594	220	1.119	28
0.735	170	1.13	27
0.76	160	1.148	26
0.79	150	1.172	25
0.82	140	1.21	24
0.846	130	1.25	23
0.87	120	1.3	22
0.9	110	1.4	21
0.915	105	1.45	20
0.925	100	1.5	19
0.935	95	1.57	18
0.953	90	1.63	17
0.96	85	1.69	16
0.975	80	1.761	15
0.985	75	1.82	14
1.004	70	1.906	13
1.01	67	1.95	12
1.017	64	2	11
1.03	61	2.05	10
1.035	57	2.117	9
1.04	55	2.17	8
1.047	52	2.24	7
1.052	49	2.31	6
1.059	46	2.38	5
1.066	43	2.45	4
1.074	40	2.536	3
1.08	37	2.582	2
1.09	34	2.598	1.4

MAGNET TEAM
Quick Info Card

* SHIELD COOLER INFO *

CTI

Compressor.. T3487PK
Control Module..T3487PJ
Adsorber..46-255952P656
Coldhead..T3487LS
Return Lines 40'............................46-255952P636
Supply Lines 40'.......................46-255952P637
Check Valve on Return Line...........46-255952P658
Static Pressure.............................230 to 240 psig
Dynamic Pressure............................60 to 90 psig

Balzer

Adsorber..46-281034P11
Coldhead (MR Max)...............................T3487LC
Coldhead (Signa).....................................T3487PP
Compressor...T3487PH
Supply Line 20'..T3487LJ
Return Line 20'..T3487LK
Supply Line 30'.......................................T3487ME
Return Line 30'..T3487MF
Supply Line 70'.......................................T3487MC
Return Line 70'.......................................T3487MD
Time Delay Relay (11 pin, Gray)........46-281034P1
Time Delay Relay (9 pin, Orange)...46-281034P16
Phase Relay.................................46-281034P22
Static Pressure.............................230 to 240 psig
Dynamic Supply Pressure............270 to 310 psig
Dynamic Return Pressure.............90 to 120 psig

Leybold

Adsorber for RW4000............................T3487MP
Adsorber for CP4000........................46-294150P24
Coldhead...T3487MR
Compressor RW4000.............................T3487MY
Compressor CP4000................................T3487RB
Supply Line 20'..T3487PC
Return Line 20'..T3487PD
Supply Line 70'.......................................T3487MZ
Return Line 70'..T3487PA
Static Pressure.............................218 to 235 psig
Dynamic Pressure........................305 to 334 psig
*Return & Supply lines are interchangeable.

Sumitomo

Coldhead 10K.................................T3487RW
Coldhead with transition flange 4K.....T3487RZ
Compressor H2O Cooled 10K..................T3487RT
Compressor H2O Cooled 4K...................T3487RT
Compressor Air Cooled 4K....................T3487RY
Adsorber..T3487RP
2nd stage indium gasket kit............2171621
Coldhead gasket / O-ring kit..............2171620
Supply Line 20M.......................................T3487RJ
Return Line 20M.....................................T3487RK
Supply Line 6M..................................T3487RL
Supply Line 6M.................................T3487RM
Static Pressure............................16.5 to 17 Bar
Dynamic Pressure.........................22.5 to 24.5 Bar

* FORMULAS / CONVERSION *

PPM = Max Hz - Min. Hz / Freq. in MHz
Tesla = .0234875 x Freq. in MHz
Frequency = 4257.6 x Gauss

Liquid Weights

Liter of LHe = Weight in pounds / 0.275
Liter of LN2 = Weight in pounds / 1.782

Singer Flow Meter

(cubic feet/hr to liters/hr)
Where R is meter reading in cubic ft and T is time in hours.

He Gas
liter = [(R2 - R1) / (T2 - T1)] x 0.042

N2 Gas
liter = [(R2 - R1) / (T2 - T1)] x 0.03835

Dwyer Flow Meter

(SCFH to liters/Hr)
He liter = Meter reading x 0.0714

Temperature

Centigrade = 5/9 (Fahrenheit - 32)
Kelvin = Centigrade + 273.16